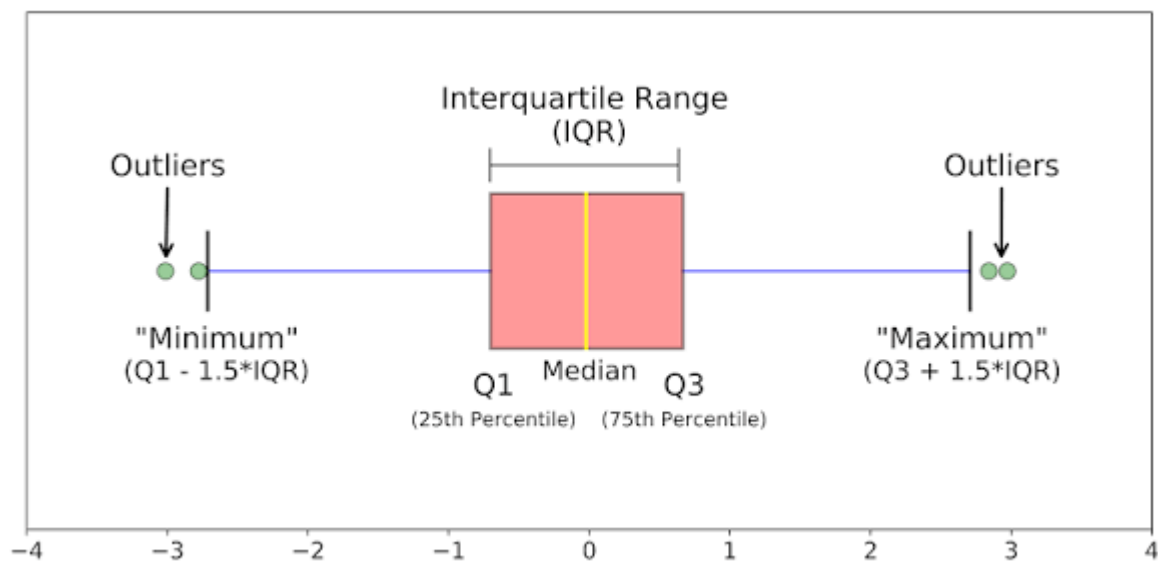


Box-Plots



$$Q1 = q_{0.25} = \frac{18.3 + 18.4}{2} = 18.35$$

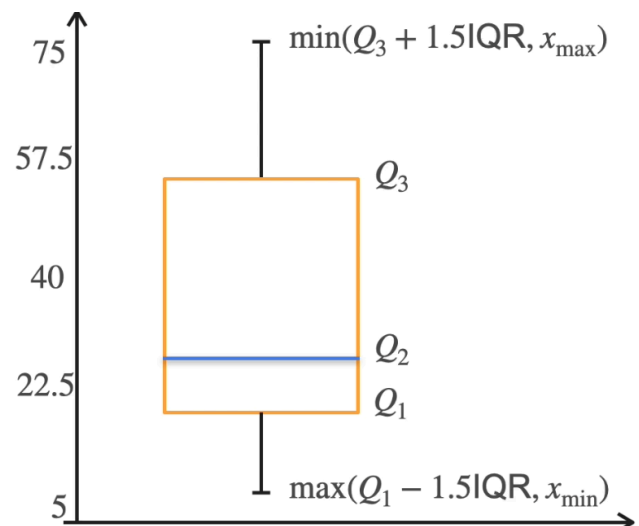
$$Q2 = q_{0.5} = \frac{25.9 + 29.7}{2} = 27.8$$

$$Q3 = q_{0.75} = \frac{51.2 + 54.7}{2} = 52.95$$

Interquartile range (IQR)

$$IQR = Q3 - Q1 = 52.95 - 18.35 = 34.6$$

$$x_{\min} = 8.7 \quad x_{\max} = 75$$



Calculation and Interpretation

1. Order the data
2. Calculate the stats
3. Draw the box
4. Mark the median
5. Add the whisker

Pros

- Identifying the outlier
- comparing the distribution
- checking the skewness
- Summarizing the large data

Cons

- **oversimplification:** it does not show the exact shape of the distribution, such as bimodal patterns
- **hidden data:** individual data points are not visible, hiding clusters or gaps